

TUTORIAL- 7

Solution 1:

$$x[n] \xleftrightarrow{DTFT} X(\Omega)$$

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(\Omega) e^{j\Omega n} d\Omega \quad \dots\dots\dots(1)$$

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n} \quad \dots\dots\dots(2)$$

Where $X(\Omega)$ is continuous and periodic (w/period 2π)

$X(\Omega)$ has a Fourier series

$$X(\Omega) \leftrightarrow a_k$$

$$X(\Omega) = \sum_{k=-\infty}^{\infty} a_k e^{-jk\Omega} d\Omega \quad (w_0 = 2\pi/2\pi = 1) \quad \dots\dots\dots(3)$$

$$a_k = \frac{1}{2\pi} \int_{2\pi} X(\Omega) e^{-jk\Omega} \quad \dots\dots\dots(4)$$

Where a_k are the Fourier series coefficient of $X(\Omega)$

Comparing equation (1) and (4)

$$x[n] = a_{-n}$$

Or

(5)

$$x[-n] = a_n$$

The original discrete signal is the reversed Fourier series of its DTFT.

Applying Parseval's Relation on $X(\Omega)$ and a_k

$$\frac{1}{2\pi} \int_{2\pi} |X(\Omega)|^2 d\Omega = \sum_{n=-\infty}^{\infty} |a_n|^2 = \sum_{n=-\infty}^{\infty} |x[n]|^2 \quad \dots\dots\dots(\text{using Eq. 5})$$

Solution 2: Consider

$$x[n] = \left(\frac{\sin(w_c n)}{\pi n} \right)^1$$

The fourier transform is as shown in the figure. Since $y[n] = (x[n])^2$ the fourier transform $Y(e^{jw})$ is

$$Y(e^{jw}) = \frac{1}{2\pi} \int X(e^{jw}) X(e^{j(w-\theta)}) d\theta$$

By defining the periodic convolution into a aperiodic signal given by

$$\hat{X}(e^{jw}) = \begin{cases} X(e^{jw}), & |w| \leq \pi \\ 0, & \text{other wise} \end{cases}$$

Hence,

$$Y(e^{jw}) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{X}(e^{jw}) X(e^{j(w-\theta)}) d\theta$$

This results in aperiodic convolution of rectangular pulse $\hat{X}(e^{j\omega})$ with the normal rectangular pulse given by $X(e^{j\omega})$. The result would be the 2nd plot in the figure given below:

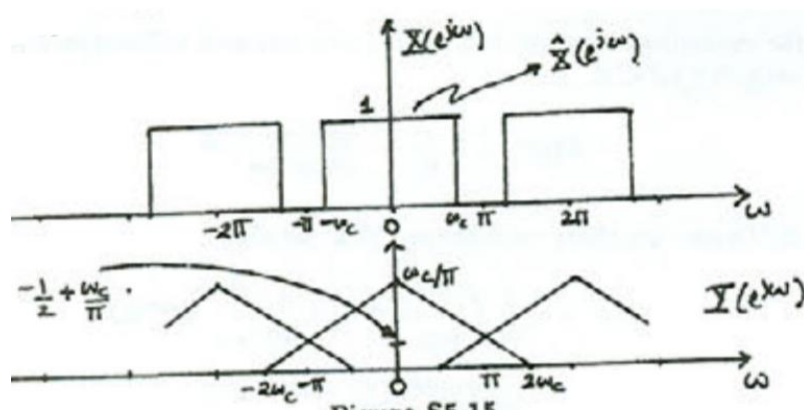
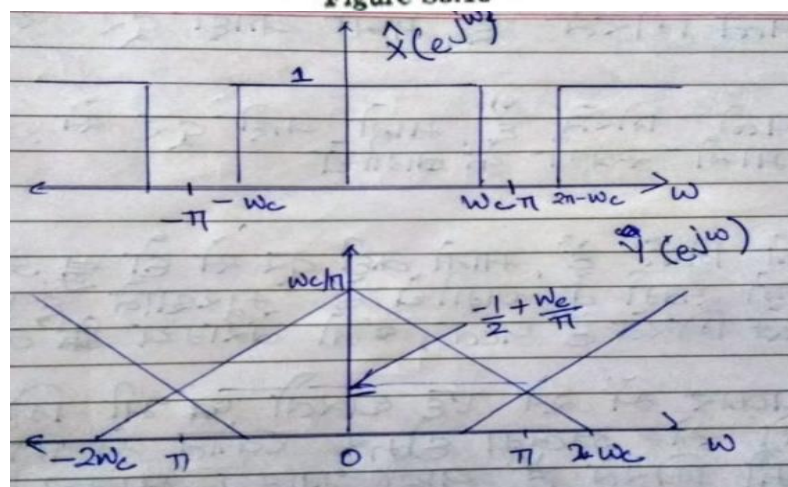


Figure S5.15



$$\text{Since } -1 + (2\omega_c/\pi) = 0.25 \text{ we get } \omega_c = \frac{5\pi}{8}$$

Solution 3:

$$\text{Let } s[n] = 0.8^n u[n]$$

$$\begin{aligned} \text{a) } x[n] &= 0.8^n u[n-2] = 0.8^2 0.8^{n-2} u[n-2] \\ &= 0.64 s[n-2] \end{aligned}$$

$$\text{then } X(\omega) = 0.64 e^{-j2\omega} S(\omega) = 0.64 \frac{e^{-j2\omega}}{1 - 0.8e^{-j\omega}}$$

$$\text{b) } x[n] = 0.8^n u[n] \cos(0.1\pi n) = \frac{1}{2} s[n] e^{j0.1\pi n} + \frac{1}{2} s[n] e^{-j0.1\pi n}$$

$$\text{which implies } X(\omega) = \frac{1}{2} S(\omega - 0.1\pi) + \frac{1}{2} S(\omega + 0.1\pi)$$

substitute for $S(\omega)$ to obtain

$$X(\omega) = \frac{1}{2} \left(\frac{1}{1 - 0.8e^{-j(\omega-0.1\pi)}} \right) + \frac{1}{2} \left(\frac{1}{1 - 0.8e^{-j(\omega+0.1\pi)}} \right)$$

c) $x[n] = n0.8^n u[n] = ns[n]$

From the properties of the DTFT:

$$\begin{aligned} X(\omega) &= j \frac{d}{d\omega} S(\omega) = j \frac{d}{d\omega} (1 - 0.8e^{-j\omega})^{-1} \\ &= -j(1 - 0.8e^{-j\omega})^{-2} (j0.8e^{-j\omega}) \end{aligned}$$

Therefore
$$X(\omega) = \frac{0.8e^{-j\omega}}{(1 - 0.8e^{-j\omega})^2}$$

d) NOTE: In problem sheet, there is typo error in part (d), it is 0.8^{-n} instead of 0.8^n

$$x[n] = 0.8^{-n} u[-n] = s[-n]$$

Then

$$\begin{aligned} X(\omega) &= \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} s[-n] e^{-j\omega n} = \sum_{n=-\infty}^{+\infty} s[n] e^{-j(-\omega)n} \\ &= S(-\omega) = \frac{1}{1 - 0.8e^{j\omega}} \end{aligned}$$

e)
$$x[n] = \begin{cases} 0.8^n & 0 \leq n \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

We can write it as

$$\begin{aligned} x[n] &= 0.8^n (u[n] - u[n - 6]) \\ &= 0.8^n u[n] - 0.8^n u[n - 6] \\ &= 0.8^n u[n] - 0.8^6 0.8^{(n-6)} u[n - 6] \\ &= s[n] - 0.8^6 s[n - 6] \end{aligned}$$

Therefore

$$\begin{aligned} X(\omega) &= 1 - 0.8^6 e^{-j6\omega} S(\omega) \\ &= \frac{1 - 0.8^6 e^{-j6\omega}}{1 - 0.8e^{-j\omega}} \end{aligned}$$

Solution 4 a) Applying the geometric series

$$X(\omega) = \sum_{n=0}^{+\infty} 0.5^n e^{-j\omega n} = \frac{1}{1-0.5 e^{-j\omega}}$$

b) Applying the geometric series

$$\begin{aligned} X(\omega) &= \sum_{n=0}^{+\infty} 0.5^n e^{-j\omega n} + \sum_{n=-1}^{-\infty} 0.5^{-n} e^{-j\omega n} = \\ &= \sum_{n=0}^{+\infty} 0.5^n e^{-j\omega n} + \sum_{n=0}^{+\infty} 0.5^n e^{j\omega n} - 1 = \frac{1}{1-0.5 e^{-j\omega}} + \frac{1}{1-0.5 e^{j\omega}} - 1 \\ &= -\frac{1.5 e^{j\omega}}{(-2 + e^{j\omega})(-0.5 + e^{j\omega})} \end{aligned}$$

c) Applying the geometric series

$$X(\omega) = \sum_{n=0}^{-\infty} 2^n e^{-j\omega n} = \sum_{n=0}^{\infty} 2^{-n} e^{j\omega n} = \frac{1}{1-0.5 e^{j\omega}}$$

d) Applying the geometric series

$$X(\omega) = \sum_{n=0}^{-\infty} 0.5^n e^{-j\omega n} = \sum_{n=0}^{\infty} 0.5^{-n} e^{j\omega n} = \sum_{n=0}^{\infty} 2^n e^{j\omega n} \rightarrow \text{does not converge}$$

and the DTFT does not exist;

e) Applying geometric series

$$X(\omega) = \sum_{n=0}^{+\infty} 2^n e^{-j\omega n} + \sum_{n=-1}^{-\infty} 2^{-n} e^{-j\omega n} = \sum_{n=0}^{\infty} 2^n e^{-j\omega n} + \sum_{n=1}^{+\infty} 2^n e^{j\omega n} \rightarrow \text{does not converge}$$

and the DTFT does not exist;

f) Expanding the cosine, we can write

$$\begin{aligned} x[n] &= \frac{3}{2} 0.8^n \times e^{j0.1\pi n} u[n] + \frac{3}{2} 1.25^n \times e^{j0.1\pi n} u[-n-1] + \\ &\quad \frac{3}{2} 0.8^n \times e^{-j0.1\pi n} u[n] + \frac{3}{2} 1.25^n \times e^{-j0.1\pi n} u[-n-1] \end{aligned}$$

which becomes

$$\begin{aligned} x[n] &= \frac{3}{2} (0.8 \times e^{j0.1\pi})^n u[n] + \frac{3}{2} (1.25 \times e^{j0.1\pi})^n u[-n-1] + \\ &\quad \frac{3}{2} (0.8 \times e^{-j0.1\pi})^n u[n] + \frac{3}{2} (1.25 \times e^{-j0.1\pi})^n u[-n-1] \end{aligned}$$

Taking the DTFT of every term, recall that

$$\text{DTFT} \{a^n u[n]\} = \sum_{n=0}^{+\infty} a^n e^{-j\omega n} = \frac{e^{j\omega}}{e^{j\omega} - a}, \text{ if } |a| < 1$$

$$\text{DTFT} \{a^n u[-n-1]\} = \sum_{n=-1}^{-\infty} a^n e^{-j\omega n} = \sum_{n=0}^{+\infty} a^{-n} e^{j\omega n} - 1 = -\frac{e^{j\omega}}{e^{j\omega} - a}, \text{ if } |a| > 1$$

Therefore we obtain

$$X(\omega) = \frac{e^{j\omega}}{e^{j\omega} - 0.8 \times e^{j0.1\pi}} + \frac{e^{j\omega}}{e^{j\omega} - 0.8 \times e^{-j0.1\pi}} - \frac{e^{j\omega}}{e^{j\omega} - 1.25 \times e^{j0.1\pi}} - \frac{e^{j\omega}}{e^{j\omega} - 1.25 \times e^{-j0.1\pi}}$$

Solution 5

$$y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = 2x[n]$$

(a) Assuming DTFT of $x[n]$ and $y[n]$ exists

$$Y(\Omega) \left[1 - \frac{3}{4}e^{-j\Omega} + \frac{1}{8}e^{-2j\Omega} \right] = 2X(\Omega)$$

$$\frac{Y(\Omega)}{X(\Omega)} = H(\Omega) = \frac{2}{1 - \frac{3}{4}e^{-j\Omega} + \frac{1}{8}e^{-2j\Omega}}$$

$$H(\Omega) = \frac{4}{1 - \frac{e^{-j\Omega}}{2}} - \frac{2}{1 - \frac{e^{-j\Omega}}{4}}$$

$$h[n] = 4 \left(\frac{1}{2} \right)^n u[n] - 2 \left(\frac{1}{4} \right)^n u[n]$$

(b) Using DTFT methods, we Assume on LTI system with zero initial condition $\leftrightarrow$ causal system.

These initial conditions $x[n]=0$ for all $n < n_0$

$$y[n]=0$$

form the auxiliary conditions

(c) Yes, unstable solution of the difference equation cannot be found using this method (as taking the DTFT assumes stability)

Solution 6

Here

$$Y(\Omega) = 2X(\Omega) + e^{-j\Omega}X(\Omega) - \frac{dX(\Omega)}{d\Omega}$$

(a) (i) The system is linear because if

$$x[n] = ax_1[n] + bx_2[n],$$

then

$$y[n] = ay_1[n] + by_2[n],$$

where $y_1[n]$ is obtained from $x_1[n]$ via the given transfer function. The similar result applies for $y_2[n]$.

(ii) The system is time-varying by the following argument.

If $x[n] \rightarrow y[n]$, does $x[n-1] \rightarrow y[n-1]$?

$$x[n-1] \xleftrightarrow{\mathcal{F}} e^{-j\Omega}X(\Omega)$$

The corresponding $Y(\Omega)$ is

$$2e^{j\Omega}X(\Omega) + e^{-j\Omega}X(\Omega)e^{-j\Omega} + je^{-j\Omega}X(\Omega) - e^{-j\Omega}\frac{dX(\Omega)}{d\Omega}$$

$$\neq e^{-j\Omega} \left[2X(\Omega) + e^{-j\Omega}X(\Omega) - \frac{dX(\Omega)}{d\Omega} \right]$$

(iii) If $x[n] = \delta[n]$, $X(\Omega) = 1$. Then

$$Y(\Omega) = 2 + e^{-j\Omega},$$

$$y[n] = 2\delta[n] + \delta[n-1]$$

Solution 7.

a. The Fourier transform $X(e^{j\omega})$ of the signal $x[n]$ is

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = \sum_{n_0}^{n_0+N-1} x[n]e^{-j\omega n}$$

Therefore

$$X\left(e^{j\frac{2\pi k}{N}}\right) = \sum_{n=n_0}^{n_0+N-1} x[n] e^{-j(2\pi/N)kn}$$

Now we can write the expression for FS coefficients of $\tilde{x}[n]$ as

$$a_k = \frac{1}{N} \sum_{\langle N \rangle} \tilde{x}[n] e^{-j(2\pi/N)kn} = \frac{1}{N} \sum_{n=n_0}^{n_0+N-1} x[n] e^{-j(2\pi/N)kn}$$

(Because $x[n] = \tilde{x}[n]$ in the range $n_0 \leq n \leq n_0 + N - 1$). Comparing the above equations, we get

$$a_k = \frac{1}{N} X(e^{j2\pi k/N})$$

b.

I. From the given $x[n]$, we can write

$$\begin{aligned} X(e^{j\omega}) &= 1 + e^{-j\omega} + e^{-2j\omega} + e^{-3j\omega} + e^{-4j\omega} \\ &= e^{-2j\omega} [e^{2j\omega} + e^{-2j\omega}] + e^{-2j\omega} [e^{j\omega} + e^{j\omega}] + e^{-2j\omega} \\ &= e^{-2j\omega} [2\cos 2\omega + 2\cos \omega + 1] \end{aligned}$$

II. We have got

$$a_k = \frac{1}{N} X(e^{j2\pi k/N})$$

$$\text{So, } a_k = \frac{1}{N} e^{-2j(2\pi k/N)} [2\cos 2(2\pi k/N) + 2\cos (2\pi k/N) + 1]$$